

ABOUT MYSELF

I received the B.E. degree in Information Engineering from City University of Hong Kong, in 2017. I am currently working at TCL AI Lab, as a research engineer and pursuing the PhD degree at the Department of Electrical Engineering, City University of Hong Kong. My research interests include multimodal learning, efficient foundation model pretraining and fine-tuning method, self-supervised learning, audio and visual representation learning, model reparameterization and federated learning.

EXPERIENCE

TCL Corporate Research (HK) Co., Ltd | TCL AI LAB

Research Engineer, May 2021 to Present

- Conduct the cutting-edge research related to multimodal learning, efficient fine-tuning and pretraining methods for foundation models, visual and audio representation learning, model reparameterization and federated learning.
- Design and develop a multimodal event detection prototype for home robots event detection, leveraging vLLM and audio event detection model.
- Design and develop a low-power, low-latency audio event detection (AED) system for always on smart home robots, achieving <20ms end-to-end latency, >95% precision, and >85% recall on domestic sound events
- Collaborate closely with researchers from Microsoft Research Asia to develop a generalized anomaly detection backbone (e.g., using weakly supervised and self-supervised learning) for different production lines to lower down the labelling cost and reducing deployment time.

Algorithm Engineer, May 2019 to April 2021

- Design and develop image enhance algorithm for smartphone cameras, with focus on image super resolution and diffraction removal for under-display cameras
- Optimize the image super resolution algorithm on TCL smartphones, achieving results with 1 second processing time.
- Patent five cutting-edge image enhancement techniques related to image super-resolution and diffraction removal for under-display camera.

TFI Digital Media Limited, Hong Kong | Technology & Research Team

Software Engineer, August 2017 to April 2019

- Design and development of video content protection and video quality enhancement algorithm
- Patent one US patent related to the image quality assessment technique.

EDUCATION

City University of Hong Kong, Department of Electrical Engineering | January 2019 to Early of 2026

- Part-time PhD in Electrical Engineering Department
- Advisor: Po Lai-Man

City University of Hong Kong, Department of Electronic Engineering | September 2014 to June 2017

- Major: BENG (Hons) in Information Engineering
- Awarded first class honours prize
- Awarded the Hong Kong Real Property Federation Scholarships in 2017 and Alco Holding Limited Scholarship in 2016
- Awarded Dean's List in 5 Semesters
- Gold Award in Information Product Design Competition in 2015/16

SELECTED PUBLICATION

- [1] **K. W. Lau**, Y. A. U. Rehman, P. P. B. de Gusmao, L.-M. Po, L. Ma, and Y. Xie, "FedRepOpt: Gradient re-parametrized optimizers in federated learning," in *Proceedings of Asian Conference on Computer Vision*, 2024, pp. 1866-1882.
- [2] **K. W. Lau**, Y. A. U. Rehman, and L.-M. Po, "AudioRepInceptionNeXt: A lightweight single-stream architecture for efficient audio recognition," *Neurocomputing*, vol. 578, p. 127432, 2024.
- [3] **K. W. Lau**, L.-M. Po, and Y. A. U. Rehman, "Large Separable Kernel Attention: Rethinking the large kernel attention design in CNN," *Expert Systems with Applications*, vol. 236, p. 121352, 2024.
- [4] T. H. Chan, **K. W. Lau**, J. Shen, G. Yin, and L. Yu, "Adaptive uncertainty estimation via high-dimensional testing on latent representations," *Advances in Neural Information Processing Systems*, vol. 36, 2024.
- [5] Y. A. U. Rehman, **K. W. Lau**, Y. Xie, L. Ma, J. Shen, "Exploring Federated Self-Supervised Learning for General Purpose Audio Understanding," *IEEE International Conference on Acoustics, Speech, and Signal Processing Workshops (ICASSPW)*, IEEE, 2024, PP. 335-340.
- [6] Y. Xie, J. Wen, **K. W. Lau**, Y. A. U. Rehman, and J. Shen, "What should be equivariant in self-supervised learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 4111-4120.

COMPETITION AWARD

- [1] 1ST Place in EPIC-Kitchens-100 2023 Challenges on EPIC-SOUNDS Audio-Based Interaction Recognition, CVPR 2023 Workshop. (Detailed report can be found on <https://arxiv.org/abs/2307.07265>)
- [2] 4th Place in UDC 2020 Challenge on Image Restoration of Under-Display Camera, CVPR 2020 Workshop. (Detailed report can be found on <https://arxiv.org/abs/2008.07742>)

PATENT

- [1] Yasar Abbas Ur Rehman, **Kin Wai Lau**, Yuyang Xie, Lan Ma, A Discriminative Framework using Vision Models for Federated Self-Supervised Audio and Image Understanding (用于多模态信息处理的数据处理系统)(申请号或专利号 : 202511667820.0) (2025)
- [2] **Kin Wai Lau**, Yasar Abbas Ur Rehman, Lai-Man Po, A lightweight single-stream architecture for efficient audio recognition (音频处理方法、装置、终端设备及计算机可读存储介质)(CN120853612A)(2024)
- [3] Yasar Abbas Ur Rehman, **Kin Wai Lau**, Yuyang Xie, Lan Ma, Federated Self-Supervised Learning for General Purpose Audio Understanding (音频数据处理方法、装置、设备和存储介质) (CN120783787A) (2024)
- [4] Ho Chun Leung, **Kin Wai Lau**, Huan Miao, Jie Liu. A super-resolution algorithm based on multi-branch high-low frequency processing (图像处理方法、装置、电子设备及计算机可读存储介质). CN115880144A (2023)
- [5] **Kin Wai Lau**, Lianping Xing, Yiqun Li, Dahai Yu. A reflection removal algorithm for under-display camera (一种图像处理方法、存储介质及终端设备). CN114387443A. (2022)
- [6] Yiqun Li, Lianping Xing, **Kin Wai Lau**, Dahai Yu. A high dynamic range deep learning algorithm for under-display camera (一种图像处理方法、存储介质及终端设备). CN114331927A (2022)
- [7] Lianping Xing, Yiqun Li, **Kin Wai Lau**, Dahai Yu. A multi-exposure fusion techniques for under-display camera. (一种图像处理方法、存储介质及终端设备). CN114283072A (2022)
- [8] Cheng Pen, **Kin Wai Lau**. A multi-frame denoising and super-resolution algorithm for smartphone camera. (一种图像处理方法及装置) CN113518243A (2021)

[9] Lai Man Po, LIU Mengyang, Yiu Fai Yuen, LI Yuming, Xuyuan Xu, Chang Zhou, Hon Wah Wong, **Kin Wai Lau**, Hon Tung Luk, Hok Kwan Cheung. Patch selection for neural network based no-reference image quality assessment. US20190362484A1 (2019)

REVIEWER

CVPR, Expert Systems with Applications, ICIP

SKILLS / CERTIFICATIONS

Technical: PyTorch, Tensorflow, Keras, Python

Spoken Language: English, Cantonese, Mandarin